

AEROLINK AIRLINES CLOUD ARCHITECTURE

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PROGRAM: DATA SCIENCE

DATA 040-007: DATA DELIVERY

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PROJECT OVERVIEW

Welcome to AeroLink, where innovation meets reliability in air travel. We connect people and places worldwide with safe, efficient, and seamless passenger services. At AeroLink, every journey is designed to be smooth and memorable. With cutting-edge flight operations and outstanding customer care, we aim to redefine travel while prioritizing sustainability and global connectivity. AeroLink is more than an airline—it's your gateway to possibilities

This report explores the cloud architecture designed for our airline and it focusing on how it supports our digital transformation and ensures high availability, security, and scalability across all departments.

MISSION

Our mission is to boost the power of cloud technology to streamline airline operations and improve passenger experiences. Through a modern, secure, and scalable cloud architecture, we aim to support innovation and ensure operational efficiency, and supports advanced analytics and machine learning, while ensuring cost-effectiveness and compliance.

OBJECTIVES

1: Real-Time Data Processing for Operational Efficiency:

Provide real time data for flight systems, customer bookings, maintenance logs, and crew schedules to enable faster, smarter decisions.

2: Cost Optimization & Performance Scalability: Optimize costs while ensuring high-performance data processing that scales with demand

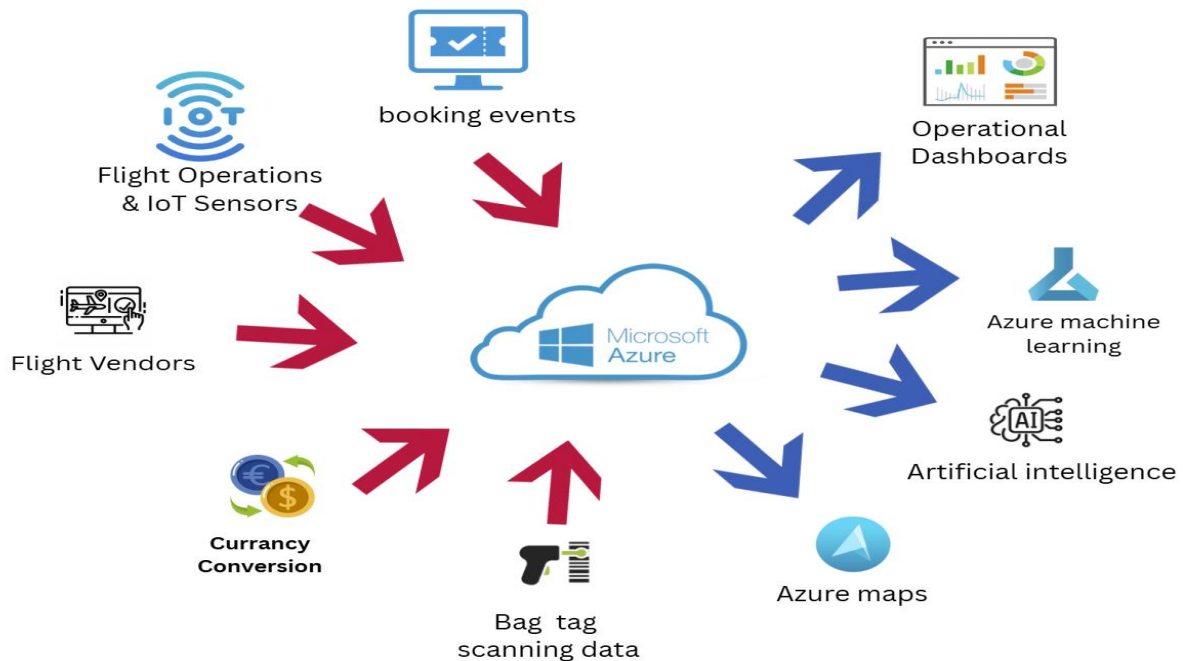
3: AI-Powered Decision Intelligence for Airline Optimization:

It helps into the decision-making process, and our airline aims to boost operational efficiency, reduce costs, and elevate the overall passenger experience by this.

4: High-Availability & Disaster Recovery with Multi-Region

Cloud Deployment: Ensure 24/7 airline operations with multi-region failover, auto-replication, and disaster recovery to prevent downtime, flight disruptions, and revenue loss

DISCOVERY PHASE



DATA SOURCES

1: Flight Operations & IoT Sensors: Collect real-time aircraft performance and environmental data.

2: Bag Tag Scanning Data: Track baggage movement and provide us real time data.

3: Booking Events: Helps us while tickets booking, cancellation and make any changes in tickets.

4: Currency Conversion: Live exchange rates for accurate international pricing and revenue tracking.

DATA DESTINATIONS

1: Operational Dashboards: It provides real time data for flight, crew, and operations teams.

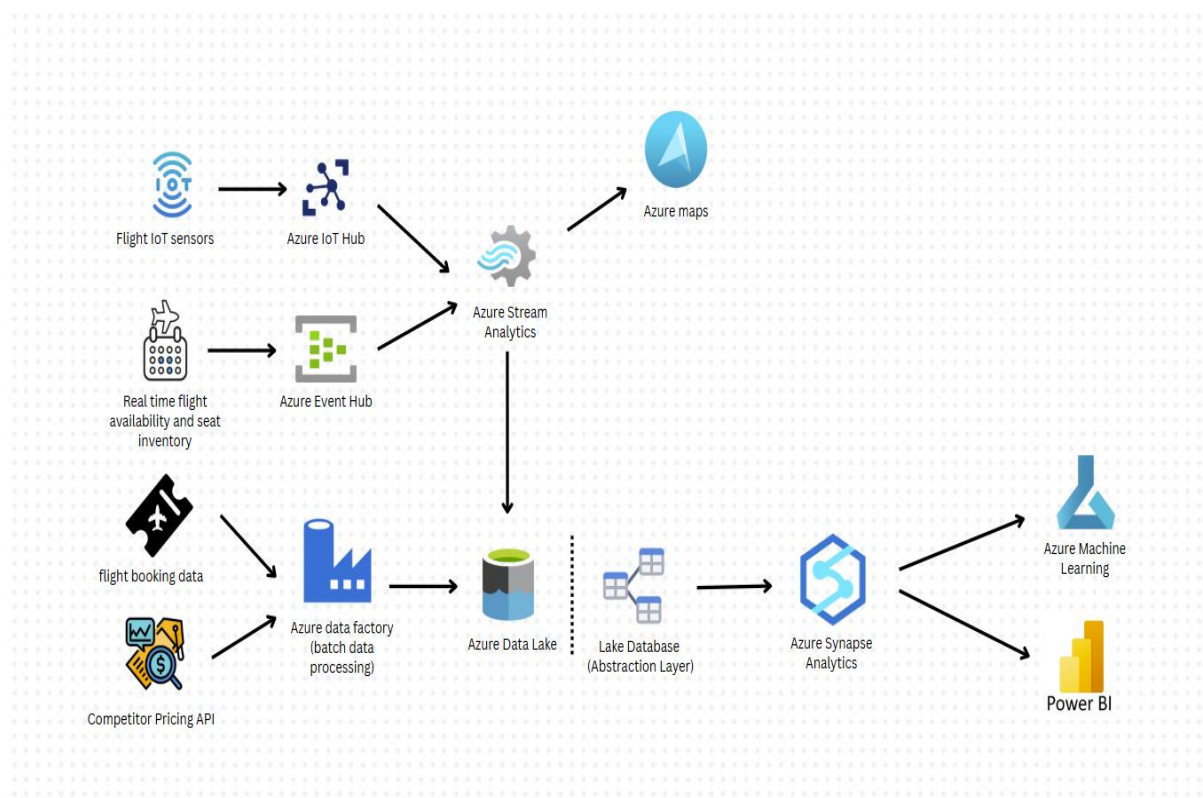
2: Azure Machine Learning: We can use it for training model like demand forecasting.

3: Azure Maps: It helps us providing flight paths and weather overlays.

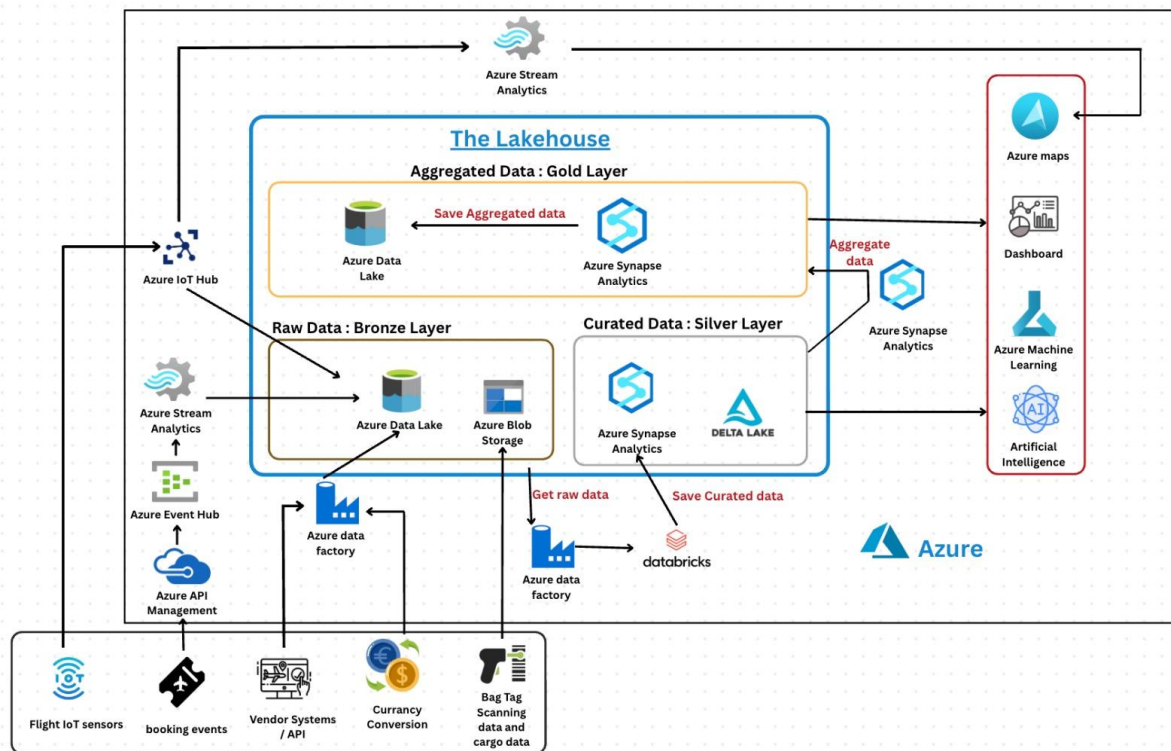
4: Artificial Intelligence: AI models powering smart operations, anomaly detection, or customer interactions.

CLOUD ARCHITECTURE

Phase-1



Final Phase



Components Involved

Azure IoT Hub: Data from aircraft sensors and IoT devices

Azure Event Hub – Real-time ingestion from booking systems, IoT sensors, and operations

Azure Data Factory – Orchestrates batch data movements and transformations

Azure Data Lake Storage Gen2 – Raw and intermediate storage

Azure Blob Storage – Staging and archiving

Azure Synapse Analytics – For querying, transformation, and analytical workloads

Power BI – For interactive dashboards and reporting

Azure API Management – Manages internal and external APIs securely and allows controlled access to services like currency conversion and vendor integrations.

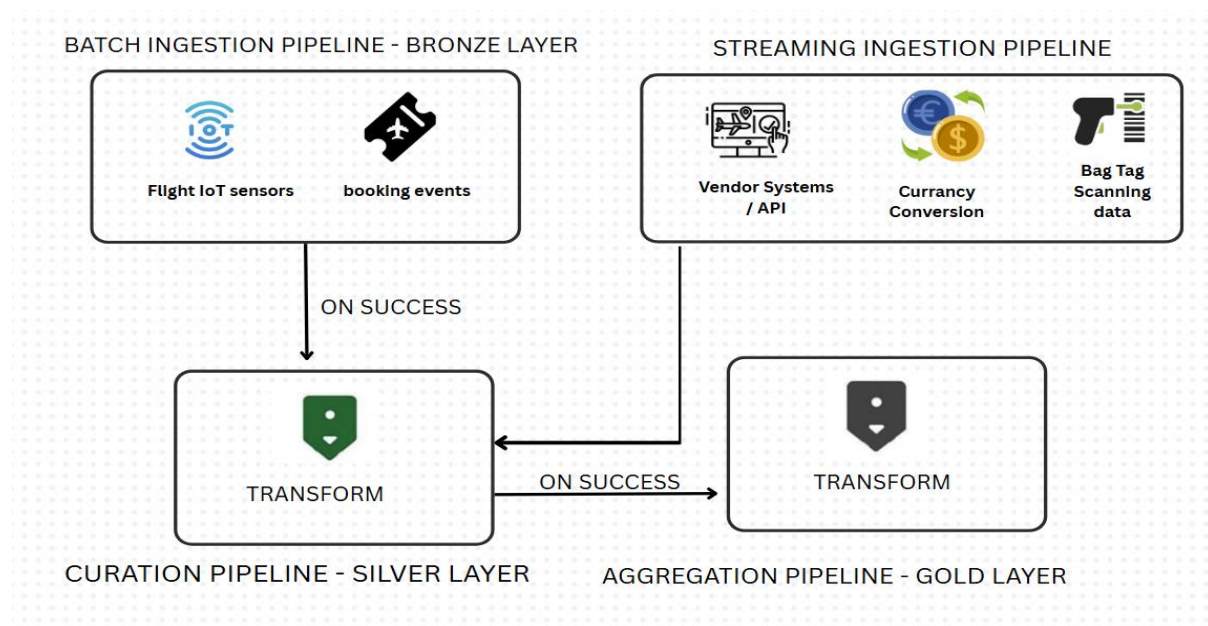
Benefits of the Architecture

Real-time Capabilities – Near real-time processing for time-sensitive operations

Cost-Optimized – Pay-as-you-go model ensures financial efficiency

Unified Storage & Analytics – Lakehouse design integrates flexibility of lakes with the performance of warehouses

DATA PIPELINE



Process Of Creating Data Pipeline

Ingestion_Layer: Purpose: The ingestion layer acts as the first entry point of the data pipeline. It captures raw data from different sources, IoT devices, CRM system and web platforms.

Data Sources:

Flight IoT Sensors (streaming)

Booking Systems, vendor system, bag tag scanning data

External APIs: Currency conversion API

Silver Layer: The silver layer is responsible for refining and curating the raw data from bronze layer. In this stage, the data undergoes significant transformation to make it structured, clean.

Activities: Join raw tables, remove bad records, standardize schemas

Gold Layer: The gold layer represents the final and most refined stage of the data pipeline, where data is transformed into high- quality.

Key Features: Business- Ready Data

Data Aggregation & Summarization

Advanced Data Modelling

Data Visualization

Pipeline Failure Handling Strategy

Failure Scenario: Data not flowing from sensors or events.

Prevention & Recovery: Set maximum retries and use DLQ (Dead Letter Queue) to catch unprocessed events

Checkpoints and Logs for resume-on-failure

Alert team if ingestion is down for a set time window (timeout).

Essence

Designing a modern cloud architecture for AeroLink Airlines using Microsoft Azure demonstrates how cloud-native solutions can revolutionize data management in the aviation industry. By embracing a scalable and intelligent data ecosystem, we are laying the groundwork for smarter operations, real-time insights, and elevated customer satisfaction.

This architecture sets the foundation for future innovations in AI, predictive maintenance, and personalized customer experiences—making AeroLink not just an airline, but a data-enabled enterprise.